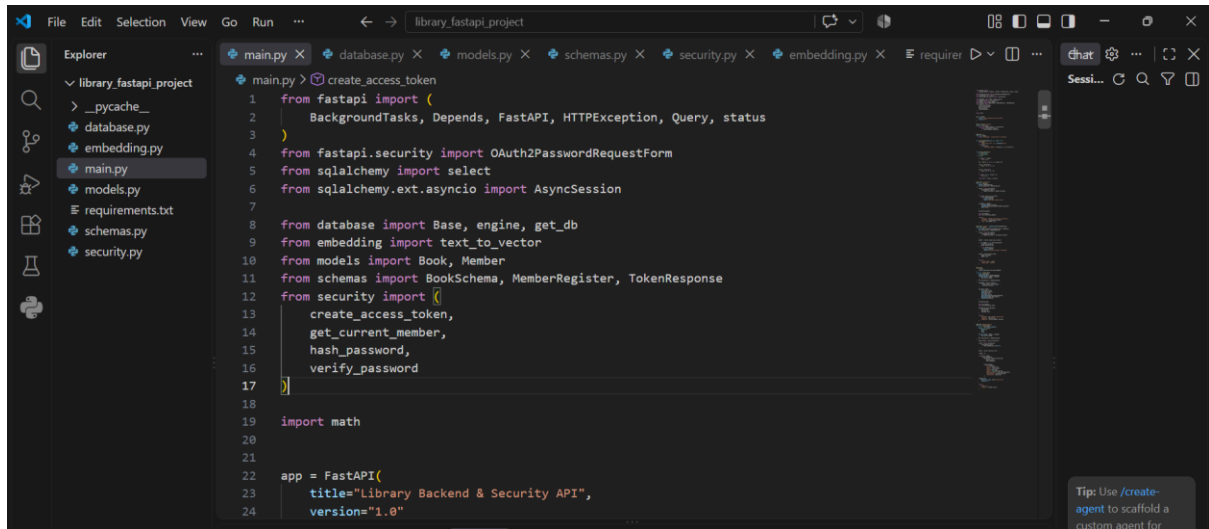
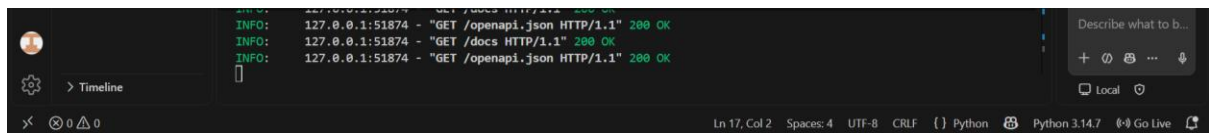


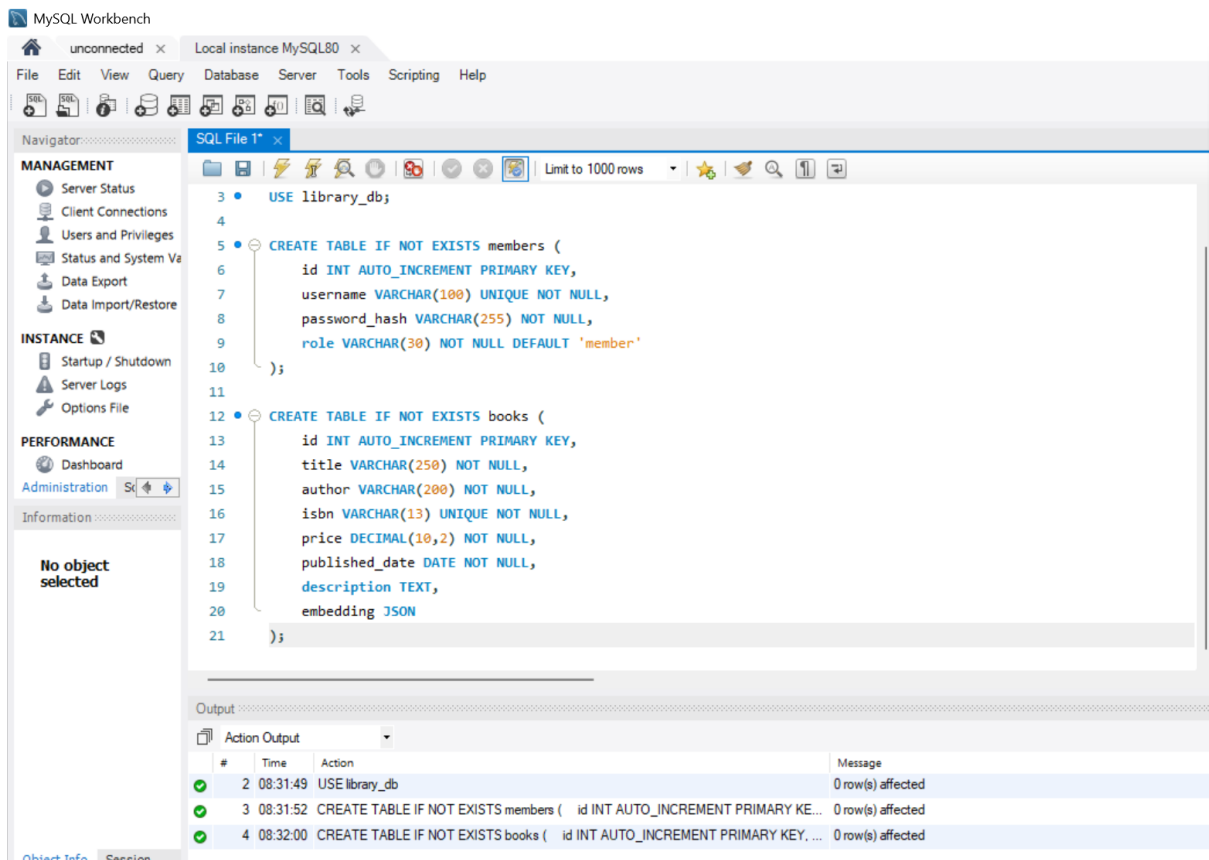
Week 5 Practical



```
1 from fastapi import (
2     BackgroundTasks, Depends, FastAPI, HTTPException, Query, status
3 )
4 from fastapi.security import OAuth2PasswordRequestForm
5 from sqlalchemy import select
6 from sqlalchemy.ext.asyncio import AsyncSession
7
8 from database import Base, engine, get_db
9 from embedding import text_to_vector
10 from models import Book, Member
11 from schemas import BookSchema, MemberRegister, TokenResponse
12 from security import (
13     create_access_token,
14     get_current_member,
15     hash_password,
16     verify_password
17 )
18
19 import math
20
21
22 app = FastAPI(
23     title="Library Backend & Security API",
24     version="1.0"
```

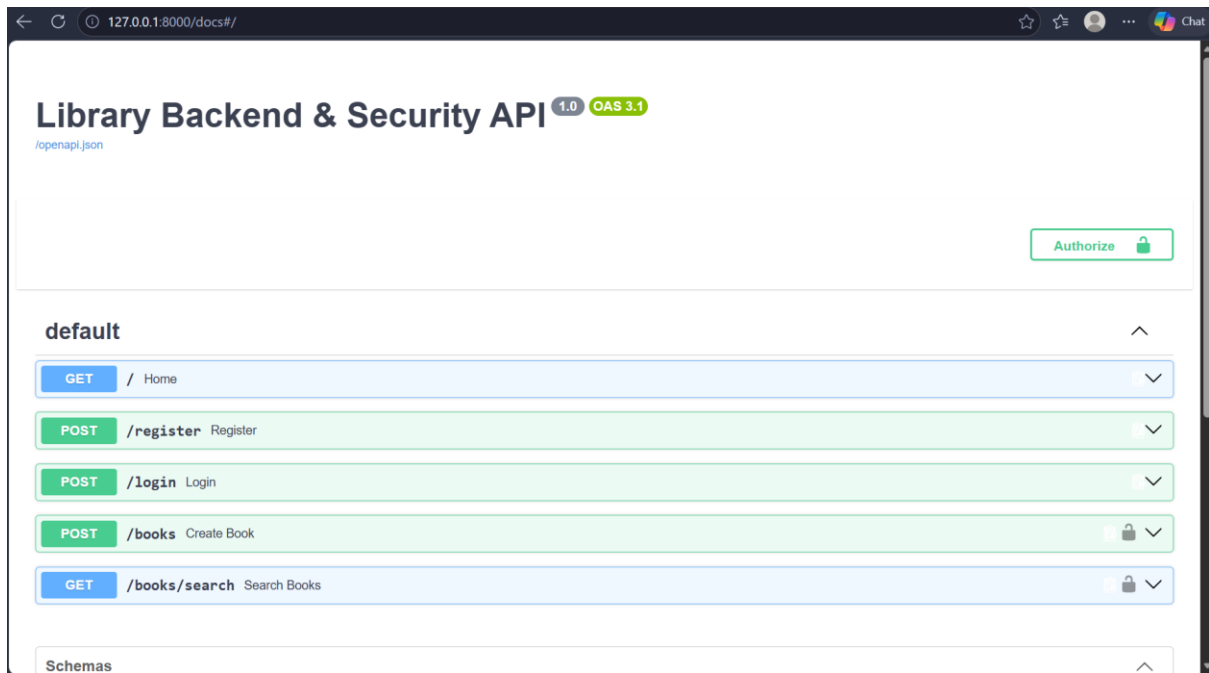


```
INFO: 127.0.0.1:51874 - "GET /openapi.json HTTP/1.1" 200 OK
INFO: 127.0.0.1:51874 - "GET /docs HTTP/1.1" 200 OK
INFO: 127.0.0.1:51874 - "GET /openapi.json HTTP/1.1" 200 OK
```



```
3 USE library_db;
4
5 CREATE TABLE IF NOT EXISTS members (
6     id INT AUTO_INCREMENT PRIMARY KEY,
7     username VARCHAR(100) UNIQUE NOT NULL,
8     password_hash VARCHAR(255) NOT NULL,
9     role VARCHAR(30) NOT NULL DEFAULT 'member'
10 );
11
12 CREATE TABLE IF NOT EXISTS books (
13     id INT AUTO_INCREMENT PRIMARY KEY,
14     title VARCHAR(250) NOT NULL,
15     author VARCHAR(200) NOT NULL,
16     isbn VARCHAR(13) UNIQUE NOT NULL,
17     price DECIMAL(10,2) NOT NULL,
18     published_date DATE NOT NULL,
19     description TEXT,
20     embedding JSON
21 );
```

#	Time	Action	Message
2	08:31:49	USE library_db	0 row(s) affected
3	08:31:52	CREATE TABLE IF NOT EXISTS members (id INT AUTO_INCREMENT PRIMARY KEY, ...	0 row(s) affected
4	08:32:00	CREATE TABLE IF NOT EXISTS books (id INT AUTO_INCREMENT PRIMARY KEY, ...	0 row(s) affected



POST/REGISTER

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/register' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "username": "student1",
    "password": "password123"
  }'
```

Request URL

http://127.0.0.1:8000/register

Server response

Code	Details
200	Response body <pre>{ "message": "Member registered successfully", "username": "student1", "role": "member" }</pre> Response headers <pre>content-length: 82 content-type: application/json date: Fri, 18 Sep 2026 03:24:57 GMT server: uvicorn</pre>

Responses

Code	Description	Links
------	-------------	-------

LOGIN

Request URL

http://127.0.0.1:8000/login

Server response

Code	Details
200	Response body <pre>{ "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ2ZmI0IjzdhVWkZw5MSIsInJvbGU0Ij0tZW1lZXIiLCJleHAiOiJlE3ODk3M0U1ODN5.1xFOUSDASyrTzioxFppeIRz4VxLYUqA627IB1Ugfg00", "token_type": "bearer"}</pre> Response headers <pre>content-length: 190 content-type: application/json date: Fri, 18 Sep 2026 03:26:22 GMT server: uvicorn</pre>

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json

Controls Accept header.

Example Value | Schema

```
{  "access_token": "string",  "token_type": "string"}
```

Library Backend & Security API 1.0 QAS 3.1

/openapi.json

default

GET / Home

POST /register Register

Parameters

No parameters

Request body required

Edit Value | Schema

Authorize

Available authorizations

Scopes are used to grant an application different levels of access to data on behalf of the end user. Each API may declare one or more scopes. API requires the following scopes. Select which ones you want to grant to Swagger UI.

OAuth2PasswordBearer (OAuth2, password)

Authorized

Token URL: /login

Flow: password

username: student1

password: *****

Client credentials location: basic

client_secret: *****

Logout Close

Cancel Reset

application/json

POST/BOOKS

http://127.0.0.1:8000/books

Server response

Code	Details
201	Response body <pre>{ "message": "Book added successfully", "book_id": 1, "added_by": "student1" }</pre> Response headers <pre>content-length: 71 content-type: application/json date: Fri, 18 Sep 2026 03:29:07 GMT server: uvicorn</pre>

Responses

Code	Description	Links
201	Successful Response	No links

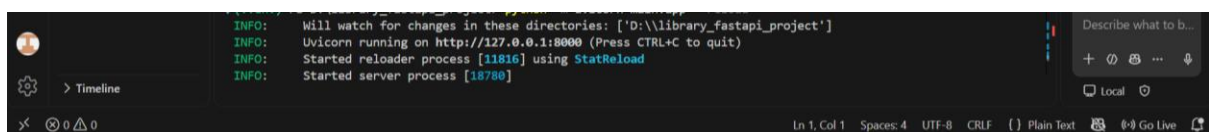
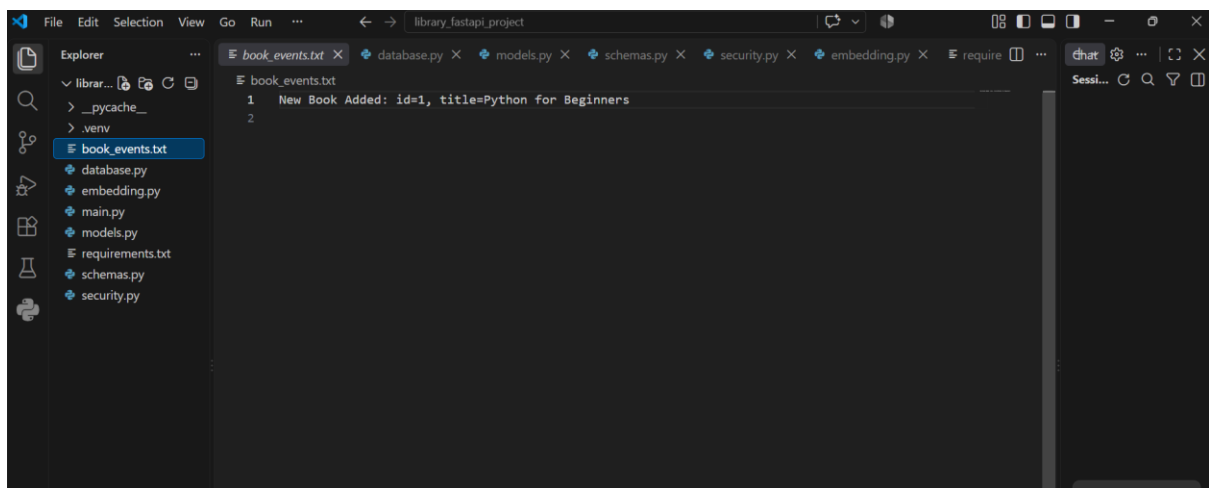
Media type

application/json

Controls Accept header.

Example Value | Schema

```
"string"
```



GET/BOOKS/SEARCH

Accept: application/json
-H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ2ZmI0IjzdhVkc2h5bGUiOiJtZW11ZXIiLCJleHAiOiJlE3ODk3MDU2ODZ9.XiBPTfGucXhQK80YdgZxNjwI-ek7gq3Yosl7ys3u

Request URL

http://127.0.0.1:8000/books/search?q=Python&limit=5

Server response

Code	Details
200	Response body <pre>{ "query": "Python", "results": [{ "id": 1, "title": "Python for Beginners", "author": "John Smith", "isbn": "1234567890123", "price": 29.99, "published_date": "2024-01-01", "description": "A beginner friendly guide to Python", "similarity": 0.0974504146403783 }]}</pre> Response headers <pre>content-length: 243 content-type: application/json date: Fri, 18 Sep 2026 03:31:20 GMT server: uvicorn</pre>

Responses

Code	Description	Links
200	Successful Response	No links

POST/BOOKS

INCORRECT ISBN:

Request URL

http://127.0.0.1:8000/books

Server response

Code	Details
422	Error: Unprocessable Content Response body <pre>{ "detail": [{ "type": "string_too_short", "loc": ["body", "isbn"], "msg": "String should have at least 13 characters", "input": "123", "ctx": { "min_length": 13 } }]}</pre> Response headers <pre>content-length: 150 content-type: application/json date: Fri, 18 Sep 2026 03:33:24 GMT server: uvicorn</pre>

Responses

Code	Description	Links
------	-------------	-------

INCORRECT PRICE:

```
    "author": "John Smith",
    "isbn": "1234567890123",
    "price": -1,
    "published_date": "2024-01-01",
    "description": "A beginner friendly guide to Python"
  }
}
```

Request URL

http://127.0.0.1:8000/books

Server response

Code	Details
422	Error: Unprocessable Content

Response body

```
{
  "detail": [
    {
      "type": "greater_than",
      "loc": [
        "body",
        "price"
      ],
      "msg": "Input should be greater than 0",
      "input": -1,
      "ctx": {
        "gt": 0
      }
    }
  ]
}
```

Response headers

```
content-length: 124
content-type: application/json
date: Fri, 18 Sep 2026 03:35:27 GMT
server: uvicorn
```

INCORRECT THAT IS FUTURE PUBLISHED DATE:

```
    "author": "John Smith",
    "isbn": "1234567890123",
    "price": 1,
    "published_date": "2030-01-01",
    "description": "A beginner friendly guide to Python"
  }
}
```

Request URL

http://127.0.0.1:8000/books

Server response

Code	Details
422	Error: Unprocessable Content

Response body

```
{
  "detail": [
    {
      "type": "value_error",
      "loc": [
        "body",
        "published_date"
      ],
      "msg": "Value error, published_date cannot be in the future",
      "input": "2030-01-01",
      "ctx": {
        "error": {}
      }
    }
  ]
}
```

Response headers

```
content-length: 167
content-type: application/json
date: Fri, 18 Sep 2026 03:36:55 GMT
server: uvicorn
```

Responses

Code	Description	Links
------	-------------	-------